



TRIBHUVAN UNIVERSITY
Faculties of Humanities and Social Sciences

AN INTERNSHIP REPORT AS
“Web Application Assistant Developer”

Submitted To:
Department of Computer Application
Kathmandu National College

In partial fulfillment of the requirements for the Bachelors in Computer Application

Submitted By:
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BCA 7th Semester, IV Year
April and 2025

Under the Supervision of
Kamal Pathak

CERTIFICATE OF COMPLETION



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INTERNSHIP CERTIFICATE

This is to certify that **Mr. Abhisekh Shakya**, a student of **Kathmandu National College, Koteshwor**, has successfully completed a **3-month internship** at **Ram Janaki Softwares Pvt Ltd.** from **January 5, 2025** to **April 5, 2025**.

During this period, he was a part of our **Web Development Team** where he received hands-on training and practical experience in **Laravel Framework Development**. He actively contributed to various project activities involving backend development, API integration, and database management for the project entitled **"Gaming Paradise"** using Laravel for website and JavaFX for software.

Abhisekh demonstrated strong technical skills, a positive attitude, and the ability to learn quickly. His performance during the internship was commendable.

We appreciate his contributions and wish him success in all future endeavors.

Internship and Boot Camps are sponsored by KG Education Institute team.

Er. Keshav Karn
CEO and Project Manager
Ram Janaki Softwares Pvt. Ltd.
Date: April 10, 2025

(Signature & Company Stamp)

Mr. Kamal Adhikari
Managing Director,
KG Education Institute,
Chuchepati-7, Chabahil.

(Signature & Company Stamp)

WEB DEVELOPMENT

SOFTWARE DEVELOPMENT

GRAPHICS DESIGNING

MENTOR'S RECOMMENDATION FROM COMPANY

I am writing to highly recommend ABHISEKH SHAKYA for any future opportunities in the field of Ram Janaki Software. As his mentor during his internship at Ram Janaki Software, I had the pleasure of working closely with him and observing his growth and development as a professional.

ABHISEKH SHAKYA consistently demonstrated a strong work ethic and dedication to his work. He was always eager to learn and took on new tasks and challenges with enthusiasm. He was an asset to our team and made valuable contributions to our projects.

In addition to his technical skills and work ethic, ABHISEKH SHAKYA is a highly personable and collaborative individual. He was able to effectively communicate and work with team members and at all levels. His positive attitude and ability to work well with others made him a pleasure to work with.

I am confident that ABHISEKH SHAKYA has a bright future in Ram Janaki Software and would not hesitate to recommend him for any future opportunities. Please do not hesitate to contact me if you have any further questions about his qualifications or potential as a professional.

Sincerely,

.....

Mr. Keshav Karn

Internship Mentor



Tribhuvan University
Faculty of Humanities and Social Sciences
Kathmandu National College

SUPERVISOR'S RECOMMENDATION

I hereby recommend that this “Internship Report” prepared under my supervision by ABHISEKH SHAKYA (TU Registration No: 6-2-752-3-2020) in partial fulfillment of the requirements for the degree of Bachelor of Computer Application is recommended for the final evaluation.

.....

SIGNATURE

Mr. Kamal Pathak

SUPERVISOR

Department of BCA

Kathmandu National College



Tribhuvan University
Faculty of Humanities and Social Sciences
Kathmandu National College

LETTER OF APPROVAL

This is to certify that this Internship report prepared by ABHISEKH SHAKYA (TU Registration No: 6-2-752-3-2020) entitled as the “**INTERNSHIP IN RAM JANAKI SOFTWARE**” for partial fulfillment of the requirements for the degree of Bachelor in Computer Application has been evaluated. In our opinion, it is satisfactory in the scope and quality as an internship report for the required degree.

..... Mr. Kamal Pathak Internship Supervisor Department of BCA Kathmandu National College	 Mr. Binod Guragain HOD/ Coordinator Department of BCA Kathmandu National College
..... Mr. Keshav Karn Internal Examiner	 Mr. Ananda KC External Examiner

ACKNOWLEDGEMENT

I would like to express my deepest gratitude to my mentor at Ram Janaki Software Pvt Ltd, Mr. Keshav Karn, for his continuous support, guidance, and constructive feedback throughout my internship. His expertise and dedication have been crucial in helping me navigate through the challenges and successfully complete my projects.

I am also thankful to the entire team at Ram Janaki Software Pvt Ltd for their cooperation and for providing me with this invaluable opportunity to gain practical experience in the field.

Additionally, I extend my sincere thanks to my academic supervisor, Kamal Pathak, for their encouragement, advice, and supervision during this internship period. This internship has been a significant learning experience, and I am grateful for the knowledge, skills, and memories I have gained.

Abhisekh Shakya

TU Registration No: 6-2-752-3-2020

TU Symbol No: 75202003

April 2025

ABSTRACT

This internship report documents my 10-week experience at Ram Janaki Software Pvt Ltd, where I contributed to the development of "Gaming Paradise", a web-based gaming platform built using the Laravel framework. The primary objectives were to gain hands-on experience in web development and to apply theoretical knowledge in a practical setting. During the internship, I learned and utilized technologies such as HTML, CSS, JavaScript, jQuery, and Laravel, focusing on creating user interfaces, implementing validations, and managing database operations. Key tasks included setting up the project environment, designing and styling login and registration forms, implementing user authentication, and integrating dynamic features using Ajax. I also explored advanced CSS techniques and briefly delved into JavaFX, broadening my technical skill set. The internship resulted in a functional gaming platform with secure user authentication and a daily bonus feature, enhancing user engagement. The internship provided valuable insights into full-stack development, teamwork, and problem-solving, significantly enhancing my academic and professional growth. This experience has equipped me with practical skills and a deeper understanding of web application development, preparing me for future challenges in the field.

Keywords: HTML, CSS, JavaScript, AJAX, JQuery, PHP, Laravel, JavaFX

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LIST OF ABBREVIATIONS

AJAX	–	Asynchronous JavaScript and XML
CSS	–	Cascading Style Sheets
CSRF	–	Cross-Site Request Forgery
HTML	–	Hyper Text Markup Language
MVC	–	Model View Controller
PHP	–	Hypertext Preprocessor
Pvt. Ltd	–	Private Limited
QA	–	Quality Assurance
SQL	–	Structured Query Language
XAMPP	–	Cross-Platform, Apache, MySQL, PHP, and Perl

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CHAPTER 1: INTRODUCTION

1.1 Introduction

During my internship at Ram Janaki Software Pvt Ltd, I had the opportunity to work on the "Gaming Paradise" project, a web-based gaming platform designed to provide users with an interactive and engaging gaming experience. The project utilized the Laravel PHP framework for backend development, while HTML, CSS, and JavaScript were employed for the frontend. My role involved assisting in the development of key features such as user authentication, including login and registration systems, as well as ensuring the application was secure, user-friendly, and scalable.

The internship provided me with hands-on experience in full-stack web development, from designing user interfaces to managing database operations and implementing server-side logic. This experience was invaluable in bridging the gap between theoretical knowledge gained from academic studies and practical application in a real-world setting. The project was structured to align with the internship's objectives of enhancing technical skills and fostering professional growth, with regular module reviews conducted by my mentor, Mr. Keshav Karn, ensuring continuous learning and improvement.

1.2 Problem Statement

The primary challenge addressed during this internship was the development of a robust and secure user authentication system for the "Gaming Paradise" platform. In the context of web-based gaming, ensuring that users can securely create accounts, log in, and manage their profiles is crucial. This involves handling sensitive user data, such as emails and passwords, and protecting against common web vulnerabilities like CSRF and SQL injection.

Additionally, the platform needed to provide a seamless user experience, with intuitive interfaces for registration and login, and real-time feedback for invalid inputs. The solution required integrating frontend technologies like HTML, CSS, and JavaScript with backend logic using the Laravel framework, ensuring that the application was both functional and maintainable. The problem was compounded by the need to ensure scalability, as the platform was designed to support a growing number of users and potentially integrate additional features like leaderboards and real-time updates.

1.3 Objectives

The primary objectives of this internship were to:

- To gain practical experience in web development technologies, including HTML, CSS, JavaScript, jQuery, and the Laravel PHP framework.
- To contribute to the development of the "Gaming Paradise" project by implementing key features such as user registration, login, and dynamic updates using Ajax.
- To ensure the security and integrity of user data by implementing validation rules and CSRF protection in the application.
- To learn about database management and Object-Relational Mapping (ORM) using Laravel's Eloquent.
- To apply theoretical knowledge from academic studies to practical scenarios, thereby bridging the gap between classroom learning and industry practices.

These objectives were aligned with the course goals of gaining practical experience and applying theoretical knowledge in a professional setting, ensuring that the internship was both educational and industry-relevant.

1.4 Scope and Limitation

The scope of this internship encompassed the following areas:

- Learning and applying web development technologies, including HTML, CSS, JavaScript, jQuery, and the Laravel PHP framework.
- Contributing to the "Gaming Paradise" project by developing user authentication features (registration and login), implementing form validations, and integrating dynamic updates using Ajax.
- Gaining hands-on experience with database management using MySQL and Laravel's Eloquent ORM.
- Understanding the full-stack development process, from frontend design to backend implementation.
- Participating in module reviews and receiving feedback to improve coding practices and problem-solving skills.

These activities were focused on enhancing technical skills and contributing to the project's development, aligning with the internship's goal of providing practical experience in computer applications and information technologies.

Limitations of existing system

The internship was subject to the following limitations:

- The duration of the internship was limited, which constrained the depth and complexity of the features that could be implemented.
- The project's scope was focused on specific features of the "Gaming Paradise" platform, such as user authentication and basic dynamic updates, rather than the entire application.
- As an assistant junior developer, my contributions were primarily focused on implementing assigned tasks under the guidance of senior developers, limiting my ability to take on more complex or independent projects.

These limitations highlight the constraints within which the internship was conducted but also provided opportunities for focused learning and skill development, ensuring a realistic assessment of the work done.

1.5 Report Organization

The report organization section provides an overview of how the internship report is structured, helping the reader understand the flow and organization of the document. This internship report is organized into the following chapters to provide a comprehensive overview of the internship experience:

Chapter 1: Introduction

This chapter introduces the internship project, outlines the problem statement, states the objectives, defines the scope and limitations, and provides an overview of the report's organization.

Chapter 2: Introduction to Organization

This chapter provides detailed information about the host organization, Ram Janaki Software Pvt. Ltd, including its organizational hierarchy, working domains, and the intern department/unit.

Chapter 3: Background Study and Literature Review

This chapter covers the fundamental theories, general concepts, and terminologies related to the internship project, as well as a review of similar projects and relevant literature.

Chapter 4: Internship Activities

This chapter describes the roles and responsibilities during the internship, the description of the project(s) involved, and the tasks/activities performed, including weekly logs.

Chapter 5: Conclusion and Learning Outcomes

This chapter summarizes the internship experience, draws conclusions from the work done, and outlines the learning outcomes and skills acquired during the internship.

Additionally, the report includes preliminary sections such as the Abstract, Acknowledgement, and Table of Contents, as well as appendices, to provide a complete and well-structured documentation of the internship. This organization ensures that the report is logical and easy to follow, allowing readers to understand the progression of the internship experience and the outcomes achieved, aligning with the syllabus's emphasis on a structured and comprehensive report.

CHAPTER 2: INTRODUCTION TO ORGANIZATION

2.1 Organization Details

Ram Janaki Software is a software development company established in Nepal, as indicated by its .np domain and pricing in Nepalese Rupees (NRS). The company specializes in creating software solutions across multiple platforms, including mobile, web, and desktop applications, and extends its services to hosting solutions. Their development philosophy emphasizes "Top Level System Design" and "Easy UI System Development," which suggests a focus on delivering user-friendly and efficient software products.

Hosting packages are also available, catering to different client needs:

Table 1: Company Hosting Packages

Plan	Price	Email Addresses	Storage	Databases	Domains	Support
Small	NRS 500/Year	3	200MB	1	1(.np)	24/7 Conditional Support
Medium	NRS 1000/Year	5	0.5GB	1	1(.np)	24/7 Updated Support
Professional	NRS 1500/Year	5	1GB	5	2(.np)	24/7 Unlimited Support
Ultimate	NRS 250/month	15	2.5GB	5	5(.np)*	24/7 Updated Support
Unlimited	NRS 300/month	15	5GB	10	2(.com)	24/7 Unlimited Support

The company's service offerings include:

- Mobile Development starting from NRS 38,000 (Ex. Tax)
- Web Development starting from NRS 8,000 (Ex. Tax)
- Desktop Development starting from NRS 12,000 (Ex. Tax)

Additional features include unlimited bandwidth, traffic, free SSL certificates, and high security, compatible with all mobile devices and modern browsers. The company also offers 24/7 support, with daily live support and system administrator's active round-the-clock, enhancing customer experience. (Ram Janaki Software Pvt. Ltd., n.d.)



Figure 1: Company Logo

Company's mission likely revolves around providing high-quality, accessible software and hosting services, given their pricing structure and emphasis on support.

2.2 Organizational Hierarchy

The organizational hierarchy of Ram Janaki Software Pvt. Ltd., as depicted in the image, presents a clear top-down structure. The hierarchy is organized systematically, ensuring a seamless flow of authority and communication across all levels. This structure effectively illustrates roles and responsibilities, fostering efficient management and coordination within the organization.

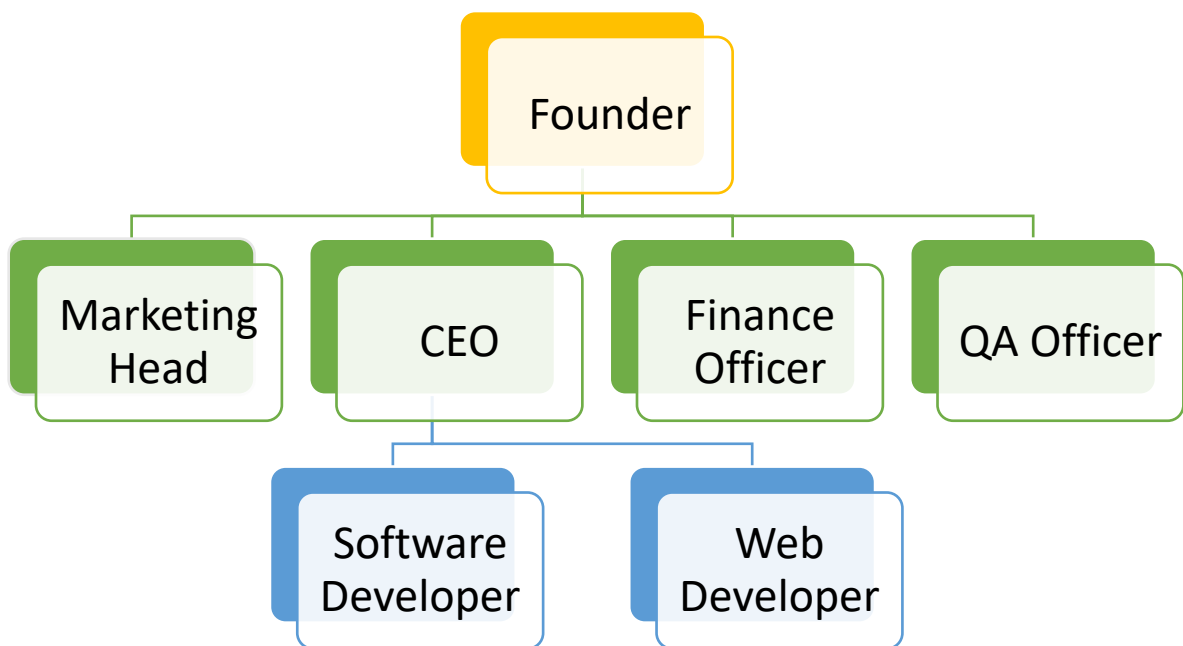


Figure 2: Organizational Hierarchy of Organization

Founder: Positioned at the apex, the Founder holds the highest authority, overseeing all aspects of the organization. This role is central to strategic decision-making and overall direction.

Direct Reports to the Founder: Four key positions report directly to the Founder:

- **CEO (Chief Executive Officer):** Responsible for managing day-to-day operations and ensuring the company's strategic goals are met. The CEO acts as a bridge between the Founder and operational teams.
- **Finance Officer:** Handles financial management, including budgeting, accounting, and financial planning, ensuring fiscal stability and compliance.

- **QA Officer:** Oversees quality control processes, ensuring that products and services meet the company's standards, likely focusing on testing and quality assurance for software development.
- **Marketing Head:** Manages marketing strategies, promotional activities, and client outreach, aiming to enhance the company's market presence and client base.

Subordinates to the CEO: Under the CEO, there are two development roles:

- **Software Developer:** Likely responsible for developing desktop applications and possibly mobile applications, given the company's service offerings.
- **Web Developer:** Focuses on creating web-based applications and websites, ensuring compatibility with modern browsers and user-friendly interfaces.

This hierarchy suggests a centralized organizational structure, with the Founder delegating authority to specialized officers and the CEO directly managing the development team. The structure aligns with a small to medium-sized software company, where the Founder maintains close oversight, and the CEO manages core operational teams, particularly development, which is central to the company's offerings.

2.3 Working Domains of Organization

Ram Janaki Software Pvt. Ltd operates in the several key domains. These domains reflect the company's expertise in software development and digital services, catering to diverse client needs. The working domains are as follows:

- **Mobile Application Development:** Focused on creating mobile applications, with packages starting at NRS 38,000 (Ex. Tax), catering to client needs for mobile solutions.
- **Web Development:** Designing and developing websites and web-based applications, starting at NRS 8,000 (Ex. Tax), emphasizing pixel-perfect icons and compatibility with modern browsers.
- **Desktop Application Development:** Building software for desktop platforms, with packages starting at NRS 12,000 (Ex. Tax), suitable for business and individual clients.
- **Hosting Services:** Providing hosting solutions with various plans (Small to Unlimited), offering features like email addresses, storage, databases, and domains, with prices ranging from NRS 500/year to NRS 300/month.

These domains highlight the company's comprehensive approach to software development and support, aligning with the needs of diverse clients in the technology sector.

2.4 Description of Intern Department/ Unit

The intern was placed in the Development Department, which is central to Ram Janaki Software Pvt Ltd.'s operations. This department is responsible for designing, developing, and maintaining software applications across mobile, web, and desktop platforms. This department is central to the company's operations, as it handles the creation of mobile, web, and desktop applications, which are core to their service offerings.

The Development Department's responsibilities include:

- Designing, coding, testing, and deploying software applications across mobile, web, and desktop platforms.
- Ensuring adherence to the company's emphasis on "Top Level System Design" and "Easy UI System Development," focusing on user-friendly interfaces and efficient system architectures.

As an intern in this department, one would typically engage in the following activities:

- Assisting with coding and programming tasks, contributing to the development of software applications under the guidance of senior developers.
- Participating in testing and debugging processes to ensure the functionality and quality of applications, identifying and resolving bugs.
- Gaining hands-on experience with real-world projects, enhancing skills in programming, problem-solving, and teamwork.

Chapter 3: Background Study and Literature Review

3.1 Background Study

The development of "Gaming Paradise" involved leveraging various web development technologies and frameworks, with a primary focus on the Laravel PHP framework. This section provides an overview of the fundamental theories, general concepts, and terminologies that are crucial for understanding the project's foundation and implementation, relating them directly to the internship activities.

Web Development Fundamentals

Web development is the process of creating websites and web applications accessible via the internet, a critical aspect of the "Gaming Paradise" project. It involves three core technologies:

- **HTML:** The standard language for structuring web pages, defining the content and layout. During the internship, HTML was used to create forms like login and registration, providing the structural foundation for user interactions.
- **CSS:** Used for styling and formatting web pages, controlling the visual appearance and layout. CSS was applied to style these forms, ensuring a user-friendly interface, enhancing the gaming platform's aesthetic appeal.
- **JavaScript:** A programming language that enables interactivity and dynamic content updates on web pages without requiring a full page reload. JavaScript, along with jQuery, was used for client-side validation and Ajax calls, such as dynamic updates for leaderboards, improving user experience in "Gaming Paradise."

PHP and Laravel Framework

PHP is a server-side scripting language widely used for web development, particularly for creating dynamic web pages, and was the backbone of Laravel in this project. Laravel is a popular open-source PHP framework that follows the MVC architectural pattern, providing a structured and efficient approach to building web applications, directly applied in "Gaming Paradise."

Key features of Laravel utilized in the project include:

- **MVC Architecture:** Separates the application into three components Models (data), Views (user interface), and Controllers (logic) enhancing code organization and maintainability. This was evident in the project's structure, with models for user data, views for forms, and controllers for logic handling.

- **Routing:** Defines how the application responds to HTTP requests, mapping URLs to specific controller actions. For example, routes were set up for /login and /register to handle user authentication, a core feature of "Gaming Paradise."
- **Eloquent ORM:** An Object-Relational Mapping system that simplifies database interactions by allowing developers to work with database tables as PHP objects. This was used to manage user data and other project-related information, such as game scores, streamlining database operations.
- **Migrations:** A version control system for the database schema, enabling easy management of database changes. Migrations were used to create and modify the users table, ensuring a flexible database structure for the gaming platform.
- **Authentication:** Laravel's built-in authentication system was leveraged to handle user registration, login, and session management, ensuring secure access to the application, a critical aspect for user accounts in "Gaming Paradise."

Database Management

Databases are essential for storing and managing data in web applications, a key component of "Gaming Paradise." The project utilized a relational database, likely MySQL, which is commonly paired with Laravel.

- **Relational Databases:** Organize data into tables with predefined relationships (e.g., one-to-many, many-to-many), ensuring data integrity and efficient querying. This was crucial for managing user relationships with games or scores in the project.
- **SQL (Structured Query Language):** The standard language for managing relational databases, used for operations like creating, reading, updating, and deleting (CRUD) data. SQL was abstracted through Eloquent ORM, but understanding it was necessary for database design.
- **Eloquent ORM:** Provides fluent interface for database interactions, abstracting complex SQL queries and simplifying data management. Eloquent was used to create and query user records, enhancing efficiency in handling game-related data.

User Authentication and Security

User authentication is critical for controlling access to web applications, directly relevant to "Gaming Paradise's" user management system. Laravel provides robust authentication features, including:

- **Session Management:** Tracks user sessions to maintain login states across pages, ensuring seamless user experience in the gaming platform.
- **CSRF Protection:** Prevents cross-site request forgery attacks by validating tokens on form submissions, such as the login and registration forms, enhancing security during user interactions.
- **Input Validation:** Ensures user inputs are sanitized and validated to prevent security vulnerabilities like SQL injection. For example, email and password fields were validated both on the client-side (using JavaScript) and server-side (using Laravel's validation rules), ensuring data integrity for user accounts.

These concepts formed the backbone of "Gaming Paradise," enabling the creation of a secure, scalable, and maintainable web application, directly reflecting the internship's technical activities.

3.2 Literature Review

The development of web-based gaming platforms has seen significant growth, with various projects leveraging modern web technologies to create engaging and interactive user experiences. This section reviews similar projects, theories, and results that are relevant to the "Gaming Paradise" project, highlighting how existing works relate to the internship project and how they inform the development process.

Similar Projects

OpenDominion

(Opendominion, n.d.) is a free online text-based multiplayer strategy game in a medieval fantasy setting, built using Laravel. It demonstrates the framework's capability in handling complex game logic, user interactions, and database management. The project showcases how Laravel's features, such as Eloquent ORM and routing, can be utilized to create a scalable and maintainable game application. Like "Gaming Paradise," OpenDominion involves user accounts, game progression, and competitive elements, providing a comparable use case for Laravel in game development, particularly in managing user data and game states, which were key activities during the internship.

OGameX

(OgameX, n.d.) is an open-source clone of the popular browser game OGame, also built with

Laravel. It highlights the use of Laravel for recreating existing game mechanics and interfaces, providing a foundation for understanding how game features can be implemented using Laravel's ecosystem. OGameX's focus on space-based strategy games aligns with the potential scope of "Gaming Paradise," especially if it includes similar gameplay elements or user interactions, such as leaderboards or game rankings, which were explored in the internship through Ajax and database queries.

Pterodactyl

(Pterodactyl, n.d.) is a free and open-source game server management panel that uses Laravel for its backend. It is designed to manage game servers at scale, offering features like server creation, configuration, and monitoring. Pterodactyl's use of Laravel underscores the framework's suitability for building robust, secure, and scalable applications in the gaming industry. If "Gaming Paradise" involves hosting or managing games, Pterodactyl provides a relevant example of how Laravel can be applied to such platforms, particularly in handling user authentication and database management, which were central to the internship project.

Theories and Results

Real-time Updates in Web Application: The tutorial on building a live game updates site with Laravel and Pusher Channels (Pusher, 2018) illustrates how real-time communication can enhance user engagement in web applications. By integrating Pusher, developers can push updates to clients instantly, which is crucial for applications requiring live data, such as sports scores or, in the context of gaming, live game stats or chat messages. This concept could be applicable to "Gaming Paradise" if it includes real-time features like live leaderboards, aligning with the internship's exploration of Ajax for dynamic updates.

Scalability and Performance: Projects like Pterodactyl emphasize the importance of scalability in gaming platforms. Handling multiple game servers and users requires efficient resource management and optimized code, areas where Laravel's architecture and community support play a significant role. For "Gaming Paradise," ensuring scalability is critical, especially if the platform is intended to support a growing number of users and games, reflecting the internship's focus on database design and Eloquent ORM for efficient data handling.

Security in Web Applications: Security is paramount in web applications, particularly those handling user data and transactions. Laravel's built-in security features, such as CSRF protection and input validation, are critical in preventing common web vulnerabilities, as demonstrated in various Laravel-based projects. These features were directly applied in "Gaming Paradise" to secure user authentication and data integrity, mirroring the internship's emphasis on secure form submissions and validation processes.

These similar projects and studies provide valuable insights into the development of "Gaming Paradise," highlighting best practices, architectural decisions, and the potential for integrating real-time features while ensuring scalability and security, directly informing the internship's technical activities and outcomes.

Chapter 4: Internship Activities

4.1 Roles and Responsibilities

During my internship at Ram Janaki Software Pvt. Ltd, I was assigned the role of an assistant junior developer, with a focus on contributing to the development of the "Gaming Paradise" project. My primary responsibilities included:

- **Learning and Applying Web Development Technologies:** I was tasked with gaining proficiency in HTML, CSS, JavaScript, jQuery, and the Laravel PHP framework, which are essential for building modern web applications. This involved understanding the structural, stylistic, and interactive aspects of web development, directly applied to creating user interfaces and dynamic features for the gaming platform.
- **Setting Up Development Environments:** I configured project folder structures and set up the development environment using tools like Composer, Xampp, and VS Code. This ensured a structured approach to file management and a functional local server for testing and development, crucial for the project's scalability.
- **Creating User Interface Components:** I designed and styled user interface elements, such as login and registration forms, using HTML, CSS, and Blade templates, ensuring a user-friendly and responsive design. This involved applying CSS techniques like positioning (absolute, relative), z-index, and animations to enhance the visual appeal and usability of the gaming platform.
- **Implementing Validation Logic:** I developed client-side validation using JavaScript and jQuery, as well as server-side validation using Laravel's built-in validation rules, to ensure data integrity and security. This included validating email formats with regex, password lengths (minimum 8, maximum 25 characters), and ensuring confirm password matches, critical for user authentication in "Gaming Paradise."
- **Developing Backend Functionality:** I created routes, controllers, and models in Laravel to handle user authentication, form submissions, and database interactions, leveraging Laravel's MVC architecture and Eloquent ORM. This involved defining routes in web.php, creating controllers like AuthController and RegisterController, and using Eloquent to manage database operations, ensuring a robust backend for the gaming application.

- **Integrating Frontend and Backend:** I ensured seamless communication between the frontend and backend to process user inputs, store data in the database, and provide dynamic updates. This included handling form submissions, implementing Ajax calls for real-time updates, and ensuring that the frontend forms interacted effectively with the backend logic, enhancing the user experience in "Gaming Paradise."
- **Participating in Code Reviews:** I attended module reviews conducted by my mentor, Keshav Karn, where my work was evaluated, and I received feedback for improvement. These reviews were crucial for identifying areas for enhancement, ensuring code quality, and aligning my work with industry standards, particularly during activities like testing and feature implementation.
- **Exploring Additional Technologies:** I briefly explored JavaFX and FXML as part of learning additional technologies, although these were not directly applied to the "Gaming Paradise" project. This exploration provided insight into desktop application development, broadening my technical skill set, though the focus remained on web development for the internship.

These responsibilities provided me with hands-on experience in full-stack web development, from designing user interfaces to managing backend logic and database operations, while also emphasizing teamwork, problem-solving, and continuous learning. The role was inferred based on the tasks performed and the organizational hierarchy, which showed development roles under the CEO, aligning with my activities in software and web development.

4.2 Weekly log

Week 1

Responsibilities:

- Learn HTML and CSS basics to understand web development fundamentals.
- Set up project folder structure for organized file management.
- Create a sample index.html with a simple form to practice HTML.
- Style the form using styles.css to enhance visual appeal.
- Link script.js to index.html for basic JavaScript practice, introducing interactivity.

Activities:

Date	Activities	Remarks
Jan 5 (Sun)	Learned basic HTML tags and document structure.	
Jan 6 (Mon)	Created index.html with a simple form for practice.	
Jan 7 (Tue)	Learned CSS selectors, properties, and styling techniques. Styled the form using styles.css, focusing on layout and design.	
Jan 8 (Wed)	Learned basic JavaScript concepts (variables, functions).	
Jan 9 (Thu)	Learned DOM Manipulation with JavaScript concepts.	
Jan 10 (Fri)	Added JavaScript to display an alert on form submission. Set up project folder structure for organized file management.	

Observations:

- HTML provides the structural foundation for web pages, while CSS is essential for styling and layout, enhancing user experience.
- JavaScript adds interactivity, such as form submission feedback, which is crucial for dynamic web applications.
- Creating a simple form helped understand how HTML, CSS, and JavaScript integrate, providing a practical learning experience.
- Organizing files into a proper folder structure improves project management.

Next Step:

- Learn AJAX, JQuery, PHP, Xampp use

Mentors Approval

Name: Mr. Keshav Karn

Signature: _____

Supervisor Approval

Name: Mr. Kamal Pathak

Signature: _____

Week 2

Responsibilities:

- Implement jQuery functionality
- Create AJAX form submissions
- Set up PHP backend processing
- Configure database connections

Activities:

Date	Activities	Remarks
Jan 12 (Sun)	Learned jQuery selectors and events, Enhanced form interactivity	
Jan 13 (Mon)	Implemented AJAX calls for form submission, Asynchronous data handling	
Jan 14 (Tue)	Implemented basic JavaScript validation, Email and password validation	
Jan 15 (Wed)	PHP Basics, Syntax, Form Handling	
Jan 16 (Thu)	Session Handling, database Connection with Xampp	
Jan 17 (Fri)	Data Persistence Implementation, Module review and MOCK by Keshav Karn	

Observations:

- jQuery simplifies DOM manipulation
- AJAX enables smooth user experience
- Understanding of server-side processing

Next Step:

- Install Laravel and set up the development environment to begin building the actual application.
- Start learning about Blade templates for dynamic view creation in Laravel.

Mentors Approval

Name: Mr. Keshav Karn

Signature: _____

Supervisor Approval

Name: Mr. Kamal Pathak

Signature: _____

Week 3

Responsibilities:

- Install Laravel using Composer to set up the framework for the project.
- Configure the development environment with Composer, Xampp, and VS Code.
- Understand the basic Laravel directory structure (app, public, resources, etc.).
- Create login.blade.php with a login form and style the form.

Activities:

Date	Activities	Remarks
Jan 19 (Sun)	Installed Composer on the local machine.	
Jan 20 (Mon)	Ran composer create-project laravel/laravel gaming-paradise-project to set up the project.	
Jan 21 (Tue)	Set up Xampp for local server and database, configured .env for database connection.	
Jan 22 (Wed)	Created resources/views/login.blade.php with a login form using Blade syntax.	
Jan 23 (Thu)	Added CSS styling to public/css/styles.css and linked login form.	
Jan 24 (Fri)	Linked styles.css to login.blade.php for consistent styling.	

Observations:

- Laravel is a robust PHP framework that provides built-in features like routing, authentication, and ORM, making web development efficient.
- The .env file is used for environment-specific configurations, such as database credentials, ensuring flexibility.

Next Step:

- Set up routes and controllers to handle login functionality, moving towards backend integration.

Mentors Approval

Name: Mr. Keshav Karn

Signature: _____

Supervisor Approval

Name: Mr. Kamal Pathak

Signature: _____

Week 4

Responsibilities:

- Set up routes in web.php for /login to define how the application responds to requests.
- Create AuthController to handle login logic, organizing backend functionality.
- Implement form submission handling to process user input securely.

Activities:

Date	Activities	Remarks
Jan 26 (Sun)	Added script.js for form validation in public/js/, focusing on email and password fields.	
Jan 27 (Mon)	Linked script.js to login.blade.php for client-side validation.	
Jan 28 (Tue)	Implemented email validation using regex in JavaScript for format checking.	
Jan 29 (Wed)	Implemented password length validation in javascript	
Jan 30 (Thu)	Defined /login GET and POST routes in web.php for form display and submission.	
Jan 31 (Fri)	Created AuthController with showLoginForm and login methods to handle requests.	

Observations:

- Client-side validation improves user experience by providing immediate feedback, reducing server load.
- Server-side validation is still necessary for security, complementing JavaScript validation.

Next Step:

- Add CSRF protection to prevent cross-site request forgery attacks.
- Begin working on registration functionality to expand user management.

Mentors Approval

Name: Mr. Keshav Karn

Signature: _____

Supervisor Approval

Name: Mr. Kamal Pathak

Signature: _____

Week 5

Responsibilities:

- Add CSRF protection to the login form to enhance security.
- Test the entire login flow to ensure it works as expected.
- Learn about database setup in Laravel for storing user data understanding persistence.
- Create migration for the users table to define the database schema.

Activities:

Date	Activities	Remarks
Feb 2 (Sun)	Added @csrf to login.blade.php for security.	
Feb 3 (Mon)	Tested login with valid and invalid credentials to verify functionality.	
Feb 4 (Tue)	Learned about Laravel migrations for database schema management.	
Feb 5 (Wed)	Created migration for users table with fields like email, password, etc.	
Feb 6 (Thu)	Ran php artisan migrate to create the users table in the database.	
Feb 7 (Fri)	Understood Eloquent ORM basics for interacting with the database using PHP objects.	

Observations:

- CSRF protection is crucial for preventing unauthorized form submissions, enhancing application security.
- Testing ensures the login flow works correctly, identifying any bugs or issues early.

Next Step:

- Implement registration functionality, creating register.blade.php and setting up routes.
- Explore additional Laravel features like validation and authentication for registration.

Mentors Approval

Name: Mr. Keshav Karn

Signature: _____

Supervisor Approval

Name: Mr. Kamal Pathak

Signature: _____

Week 6

Responsibilities:

- Create register.blade.php with a registration form to expand user management.
- Style the form using CSS for a consistent user interface.
- Add JavaScript validation for name, email, password, and confirm password to ensure data integrity.

Activities:

Date	Activities	Remarks
Feb 9 (Sun)	Created register.blade.php with form fields for registration.	
Feb 10 (Mon)	Added CSS styling to register.blade.php using public/css/style.css	
Feb 11 (Tue)	Added JavaScript validation for name (not empty).	
Feb 12 (Wed)	Added email validation using regex for format checking.	
Feb 13 (Thu)	Added password length validation (min 8, max 25 characters).	
Feb 14 (Fri)	Added confirm password matching validation to ensure consistency.	

Observations:

- Reusing skills from login form creation accelerated development of the registration form.
- Additional fields like name and confirm password require specific validation logic for accuracy.
- Client-side validation enhances user experience by catching errors before submission, improving data quality.

Next Step:

- Set up routes and controllers for registration in Laravel.
- Handle user registration in the backend, saving data to the database.

Mentors Approval

Name: Mr. Keshav Karn

Signature: _____

Supervisor Approval

Name: Mr. Kamal Pathak

Signature: _____

Week 7

Responsibilities:

- Define /register GET and POST routes in web.php for registration functionality.
- Create RegisterController to handle registration logic.
- Implement user creation logic with validation to ensure data integrity.
- Save new user data to the database using Eloquent ORM.

Activities:

Date	Activities	Remarks
Feb 16 (Sun)	Defined /register routes in web.php for form display and submission.	
Feb 17 (Mon)	Created RegisterController with showRegistrationForm and store methods.	
Feb 18 (Tue)	Implemented store method with validation rules (e.g., unique email).	
Feb 19 (Wed)	Used Eloquent's create method to save new user to the database.	
Feb 20 (Thu)	Added @csrf to register.blade.php for security.	
Feb 21 (Fri)	Tested registration with different inputs to verify functionality.	

Observations:

- Laravel's validation methods ensure data integrity before processing, such as checking for unique emails.
- Testing ensures the registration process works correctly under various conditions, identifying potential issues.

Next Step:

- Learn about file uploads in PHP/Laravel for handling user profiles or game assets.
- Implement image upload features and apply advanced CSS techniques for the game interface.

Mentors Approval

Name: Mr. Keshav Karn

Signature: _____

Supervisor Approval

Name: Mr. Kamal Pathak

Signature: _____

Week 8

Responsibilities:

- Learn about file uploads in PHP/Laravel to handle multimedia content.
- Implement image upload for user profiles or game assets, enhancing the gaming experience.
- Learn advanced CSS techniques (positioning, z-index, etc.) for complex layouts.
- Work on gaming project features, integrating all learned skills.

Activities:

Date	Activities	Remarks
Feb 23 (Sun)	Gaming Architecture Design, Structure Planning	
Feb 24 (Mon)	Created game models and migrations, Database Design	
Feb 25 (Tue)	Added image upload feature for user profiles in gaming project.	
Feb 26 (Wed)	Learned advanced CSS properties (positioning: absolute, relative; z-index for layering).	
Feb 27 (Thu)	Applied CSS techniques to create a responsive game interface.	
Feb 28 (Fri)	Worked on gaming project-Bonus Feature (24 hr. refresh)	

Observations:

- File uploads can be handled securely using PHP's built-in functions and Laravel's storage system.
- Image uploads enhance user engagement, such as adding profile picture for gamers.
- Advanced CSS properties like positioning and z-index are crucial for creating complex, layered layouts.
- Building the gaming project integrated all learned skills, from frontend to backend, ensuring a cohesive application.

Next Step:

- Prepare for module review and MOCK by Keshav Karn to assess learning outcomes.

Mentors Approval

Name: Mr. Keshav Karn

Signature: _____

Supervisor Approval

Name: Mr. Kamal Pathak

Signature: _____

Week 9

Responsibilities:

- Implement multiple games.
- Point Incrementing System and testing

Activities:

Date	Activities	Remarks
Mar 2 (Sun)	Developed Tic Tac Toe game using minmax algorithm	
Mar 3 (Mon)	Created and added Snake Game using logic	
Mar 4 (Tue)	Created and added Puzzle Game using Puzzle Algorithm	
Mar 5 (Wed)	Implemented Point incrementing system, Points Tracking in DB.	
Mar 6 (Thu)	Improved UI/UX design and visual improvement and testing games	
Mar 7 (Fri)	Module review and MOCK by Keshav Karn	

Observations:

- Game Logic Complexity.
- User experience priorities.
- Score/ Points Tracking in database.

Next Step:

- Learn JavaFX and FXML as additional skills.

Mentors Approval

Name: Mr. Keshav Karn

Signature: _____

Supervisor Approval

Name: Mr. Kamal Pathak

Signature: _____

Week 10

Responsibilities:

- Learn JavaFX and FXML for desktop application development.
- Set up Netbeans and Scene Builder.
- Explore JavaFX basics and UI design.

Activities:

Date	Activities	Remarks
Mar 09 (Sun)	Installed Netbeans and set up a JavaFX project.	
Mar 10 (Mon)	Learned JavaFX basics (Stage, Scene, Elements).	
Mar 11 (Tue)	Used FXML to design UI components.	
Mar 12 (Wed)	Integrated FXML with Java code for interactivity.	
Mar 14 (Fri)	Explored Scene Builder for visual UI design.	

Observations:

- JavaFX is powerful for building desktop applications with rich interfaces.
- FXML allows separating UI design from logic.
- Scene Builder simplifies visual UI design.

Next Step:

- Finalize the "Gaming Paradise" project and prepare for module review.

Mentors Approval

Name: Mr. Keshav Karn

Signature: _____

Supervisor Approval

Name: Mr. Kamal Pathak

Signature: _____

4.3 Description of the Project(s) Involved During Internship

The primary project I worked on during my internship was "Gaming Paradise," a web-based gaming platform developed using the Laravel framework. The project aimed to create an interactive platform where users could register, log in, and engage in various gaming activities. Below is a detailed description of the project:

Project Overview: "Gaming Paradise" is a full-stack web application designed to provide users with a platform for gaming-related activities. It includes features such as user authentication, profile management, and potentially real-time updates for enhanced user engagement. The project was developed to be scalable and maintainable, leveraging Laravel's MVC architecture and built-in features like migrations and authentication support.

Objectives:

- To develop a secure and user-friendly login and registration system, ensuring users could access the platform with validated credentials.
- To manage user data and game-related information using a relational database, such as MySQL, with Eloquent ORM for efficient data handling.
- To allow users to upload profile pictures, improving personalization and engagement.
- To ensure the application is scalable and maintainable, using Laravel's routing, authentication, and ORM features to support future expansions.

Technologies Used:

- **Frontend:** HTML for structure, CSS (including advanced techniques like positioning, z-index, and animations) for styling, JavaScript and jQuery for interactivity and Ajax calls.
- **Backend:** Laravel (PHP framework) for server-side logic, Eloquent ORM for database management, and Blade templates for dynamic views.
- **Database:** MySQL (via Xampp) for storing user data, game scores, and other project-related information.
- **Additional Tools:** Composer for dependency management, VS Code for development, and Xampp for local server and database setup.

Key Features:

- **User Authentication:** A robust system for user registration and login, with CSRF protection and input validation, ensuring secure access to the platform.

- **Database Management:** Using Laravel's migrations to create and manage tables like users, and Eloquent ORM to store and retrieve data, such as user profiles and game scores.
- **Dynamic Updates:** Implementing Ajax for real-time features, such as fetching leaderboard data dynamically, improving user engagement and reducing server load.
- **File Uploads:** Allowing users to upload images for their profiles or game assets, handled securely using Laravel's file storage system.
- **Advanced Styling:** Applying CSS techniques like positioning (absolute, relative), z-index for layering, and animations to create an engaging and responsive user interface, enhancing the gaming experience.

Contribution: As an assistant junior developer, I contributed to all aspects of the project, from setting up the initial project structure to implementing key features like user authentication, database setup, and dynamic updates. My work focused on ensuring that the frontend and backend were seamlessly integrated, providing a cohesive user experience. This included creating and styling forms, implementing validations, handling database operations, and participating in module reviews to refine the application.

The "Gaming Paradise" project was designed to be scalable and maintainable, leveraging Laravel's built-in features like routing, authentication support, and Eloquent ORM. My involvement in this project allowed me to apply theoretical knowledge to practical scenarios, enhancing my understanding of web development principles and best practices, and aligning with the internship's objective of providing industry-relevant experience.

4.4 Tasks / Activities Performed

Throughout the 10-week internship, I performed a variety of tasks that contributed to the development of the "Gaming Paradise" project. Below is a detailed account of the activities I undertook each week, grouped by week for clarity:

- **Development Environment Setup:** I configured Laravel with Composer, set up Xampp for local server and database management, and used VS Code as my IDE to establish an efficient workflow.
- **Frontend Development:** I designed and styled user interfaces for login and registration forms using HTML, CSS, and Laravel's Blade templates. I also added client-side validation with JavaScript and jQuery to improve user experience.

- **Backend Development:** I built the backend for user authentication in Laravel, creating routes, controllers, and models. This included processing form submissions and managing database interactions securely with Eloquent ORM.
- **Database Management:** I designed the database schema using Laravel migrations, setting up tables for users and game data, and used Eloquent ORM to streamline database operations.
- **Dynamic Features:** I implemented Ajax to enable real-time updates, like a daily bonus feature where users could claim points every 24 hours without refreshing the page, enhancing interactivity.
- **Security Measures:** I incorporated CSRF protection in forms and applied both client-side and server-side validation to protect against threats like SQL injection and cross-site request forgery.
- **Technology Exploration:** In the final week, I explored JavaFX and FXML with Scene Builder for desktop UI development, which expanded my knowledge, though it wasn't part of the main project.

These tasks provided me with a comprehensive understanding of the full-stack development process, from frontend design to backend implementation. They equipped me with practical skills in using Laravel for web application development, including routing, database management, validation, and dynamic updates. Additionally, I gained experience in teamwork, problem-solving, and receiving constructive feedback during module reviews, ensuring alignment with industry standards and the internship's objectives.

Chapter 5: Conclusion and Learning Outcomes

5.1 Conclusion

During my internship, I underwent comprehensive training in full stack web development, focusing on both frontend and backend technologies. The journey started with basic web fundamentals and progressed to advanced Laravel framework implementation, culminating in the development of Gaming Paradise platform. The internship provided structured learning through various modules:

1. Authentication System Development

- Mastered form handling and validation
- Implemented secure login and registration systems
- Learned best practices for user authentication
- Gained experience with CSRF protection and security measures

2. Frontend Development Skills

- Advanced CSS positioning and layouts
- Interactive UI elements using jQuery and JavaScript
- Form validation using both client and server-side approaches
- Responsive design implementation

3. Backend Development Expertise

- Laravel framework implementation
- Database design and management
- MVC architecture understanding

4. Full Stack Integration

- AJAX implementation for seamless user experience
- File handling and image uploads
- Session and cookie management

The culmination of this learning was demonstrated in the Gaming Paradise project, which incorporated all these elements into a cohesive gaming platform.

5.2 Learning Outcome

Throughout my internship, I acquired a range of technical and professional skills that have greatly contributed to my growth as a developer. On the technical side, I became proficient in using Laravel for backend development, understanding its MVC architecture, routing, and Eloquent ORM. I also strengthened my front-end skills with HTML, CSS, and JavaScript, learning to create responsive and interactive user interfaces. Additionally, I gained experience with Ajax for dynamic content updates, such as implementing the daily bonus feature, and file uploads for user profile images, enhancing the personalization of the gaming experience.

Beyond technical skills, I developed important professional competencies. I improved my problem-solving abilities by tackling real-world challenges, such as debugging code and optimizing performance. Working as part of a team under the guidance of my mentor, Mr. Keshav Karn, I learned the importance of communication and collaboration in achieving project goals. I also gained insights into the software development lifecycle, from requirement gathering to testing and deployment, and understood the importance of security in web applications, particularly in handling user data and preventing vulnerabilities like CSRF attacks.

One of the most valuable lessons was recognizing the role of developers in creating safe and reliable applications, which has deepened my appreciation for ethical and secure coding practices. Additionally, exploring JavaFX and FXML, though not directly applied to the project, broadened my understanding of desktop application development, providing a foundation for future learning.

Overall, this internship has been a transformative experience, bridging the gap between academic learning and industry practice. The skills and insights gained have not only enhanced my technical expertise but also prepared me for a career in computer applications and information technologies, where I can contribute meaningfully to the development of innovative solutions.

Key Achievements

1. Successfully implemented a complete authentication system
2. Developed responsive and interactive user interfaces
3. Created efficient database structures and queries
4. Developed the Gaming Paradise platform incorporating all learned skills

The internship provided not just theoretical knowledge but practical implementation experience through hands on development of real-world applications. The regular module reviews and MOCK sessions with Keshav helped ensure the quality of learning and implementation. This experience has laid a strong foundation for my career in web development, particularly in Laravel-based applications. The culmination of this learning in the Gaming Paradise project demonstrated my ability to integrate various technologies and implement complex features in a production environment. This internship has significantly enhanced my technical capabilities and prepared me for future challenges in the field of web development.

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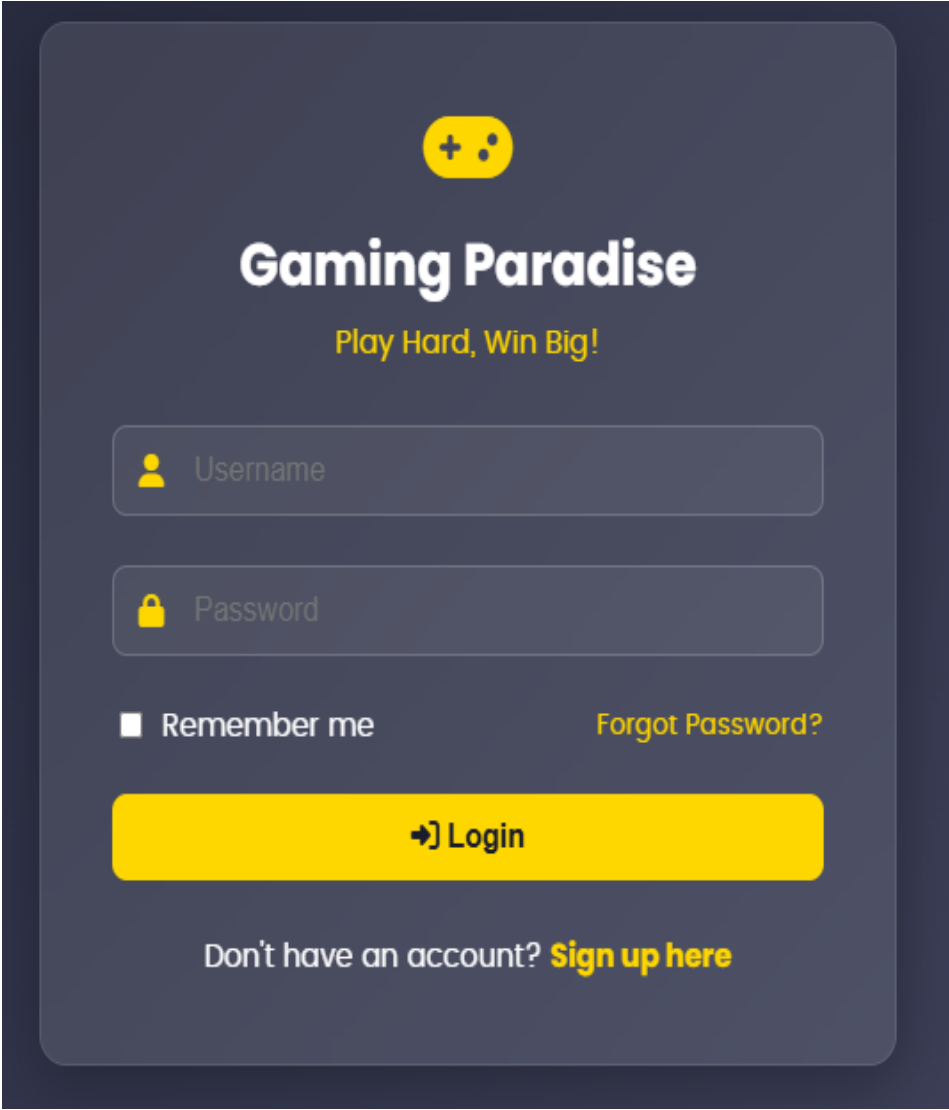
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APPENDICES



The image shows a login page for 'Gaming Paradise'. At the top is a yellow game controller icon. Below it is the title 'Gaming Paradise' in white, followed by the tagline 'Play Hard, Win Big!' in yellow. There are two input fields: 'Username' with a person icon and 'Password' with a lock icon. Below the password field is a 'Remember me' checkbox and a 'Forgot Password?' link. A large yellow 'Login' button with a right arrow icon is centered. At the bottom, it says 'Don't have an account? Sign up here'.



Gaming Paradise

Play Hard, Win Big!

 Username

 Password

☐ Remember me [Forgot Password?](#)

 Login

Don't have an account? [Sign up here](#)

Figure 3: Login Page



Create Account

Join our gaming community!

 Username

 Email

 Phone (10 digits)

☐ ♂ Male ☐ ♀ Female ☐ ○ Other

 Password

At least 8 characters
One uppercase letter
One lowercase letter
One number
One special character

 Create Account

Already have an account? [Sign in](#)

Figure 4: Register Page

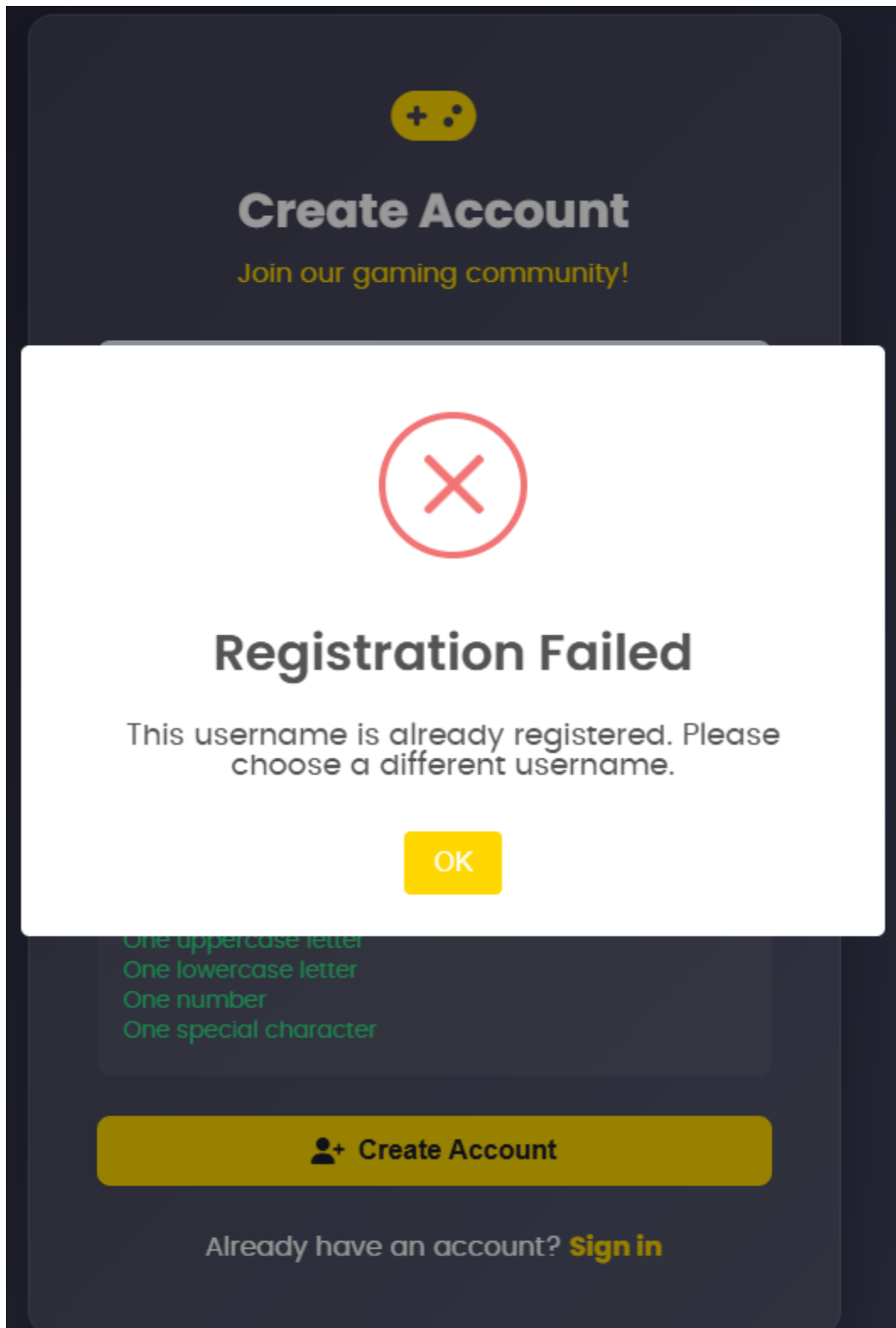


Figure 5: Testing Registering Duplicate Data

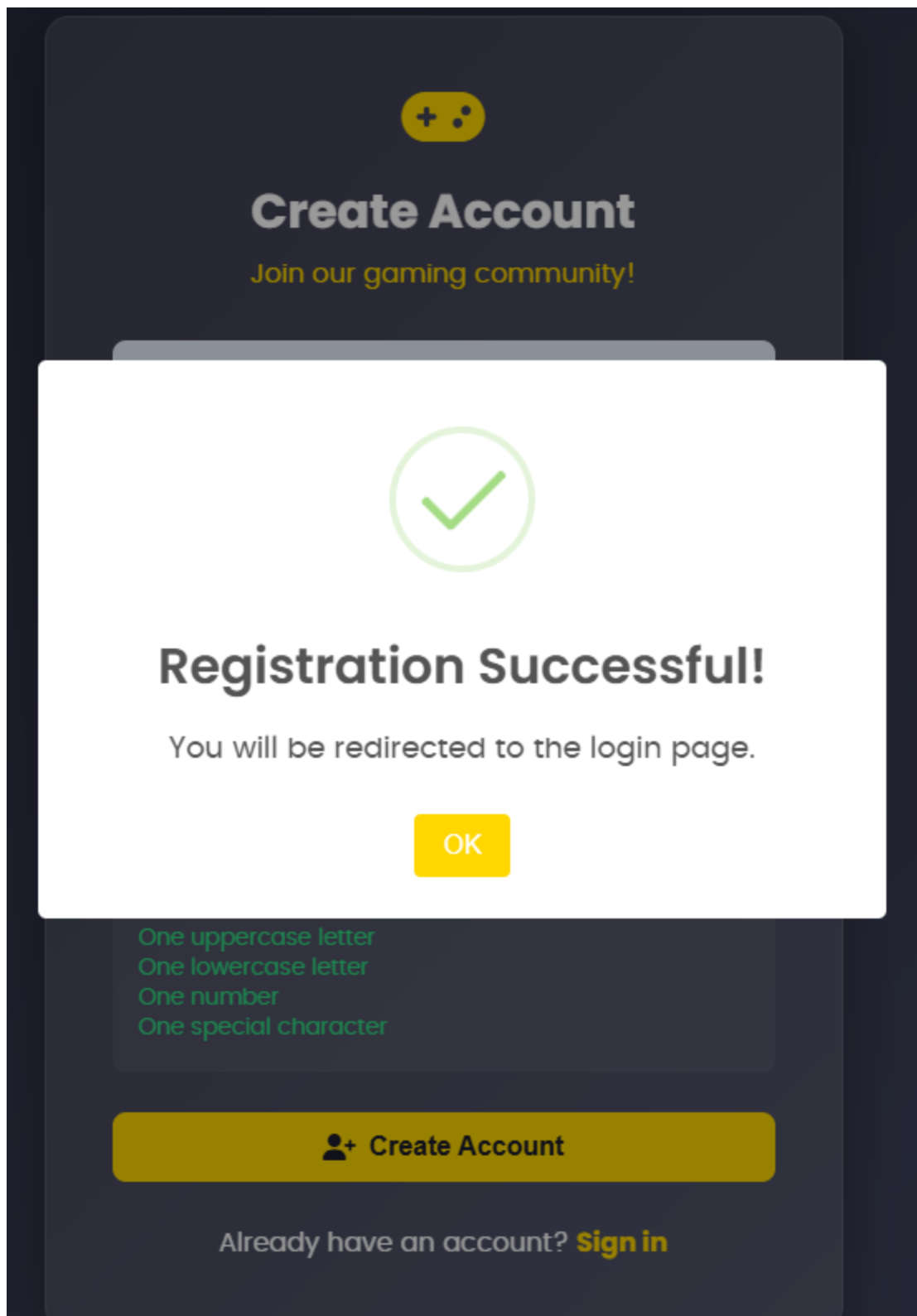


Figure 6: Registration Test Successful

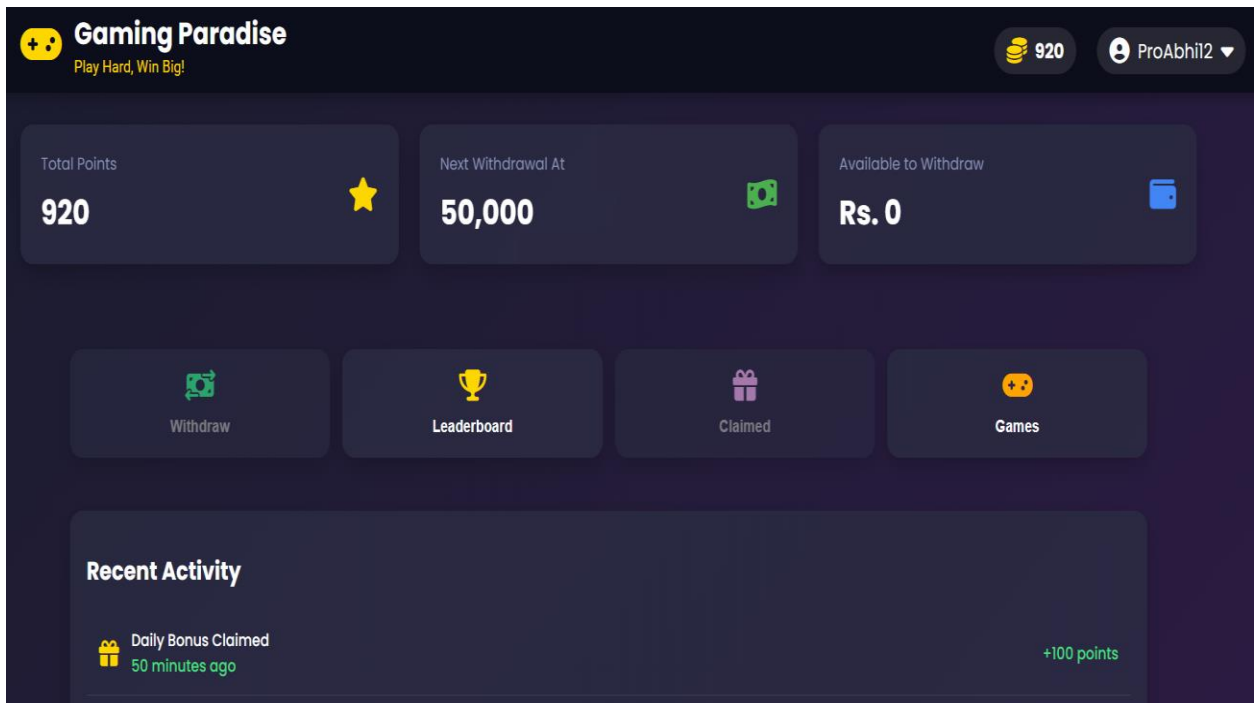


Figure 7: Dashboard Page

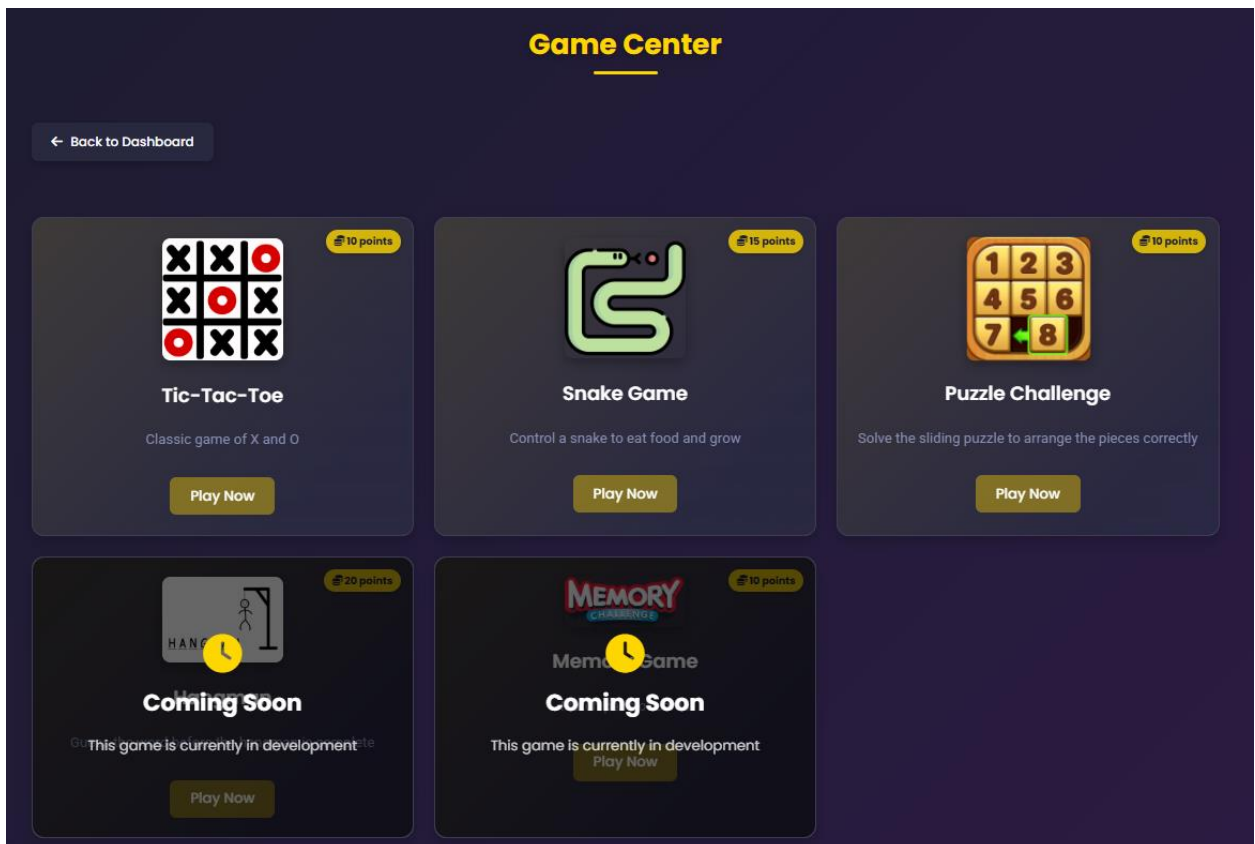


Figure 8: Game Center Page

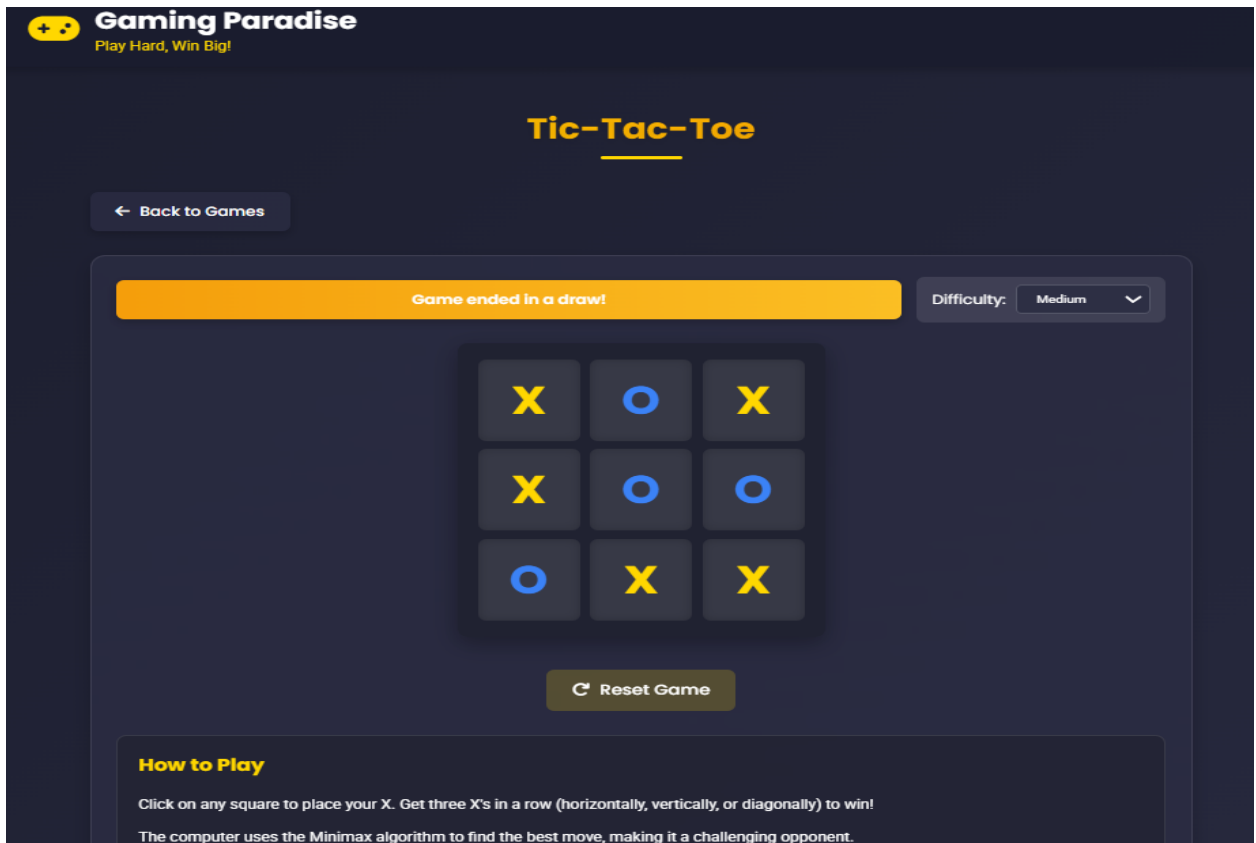


Figure 9: Tic Tac Toe Game

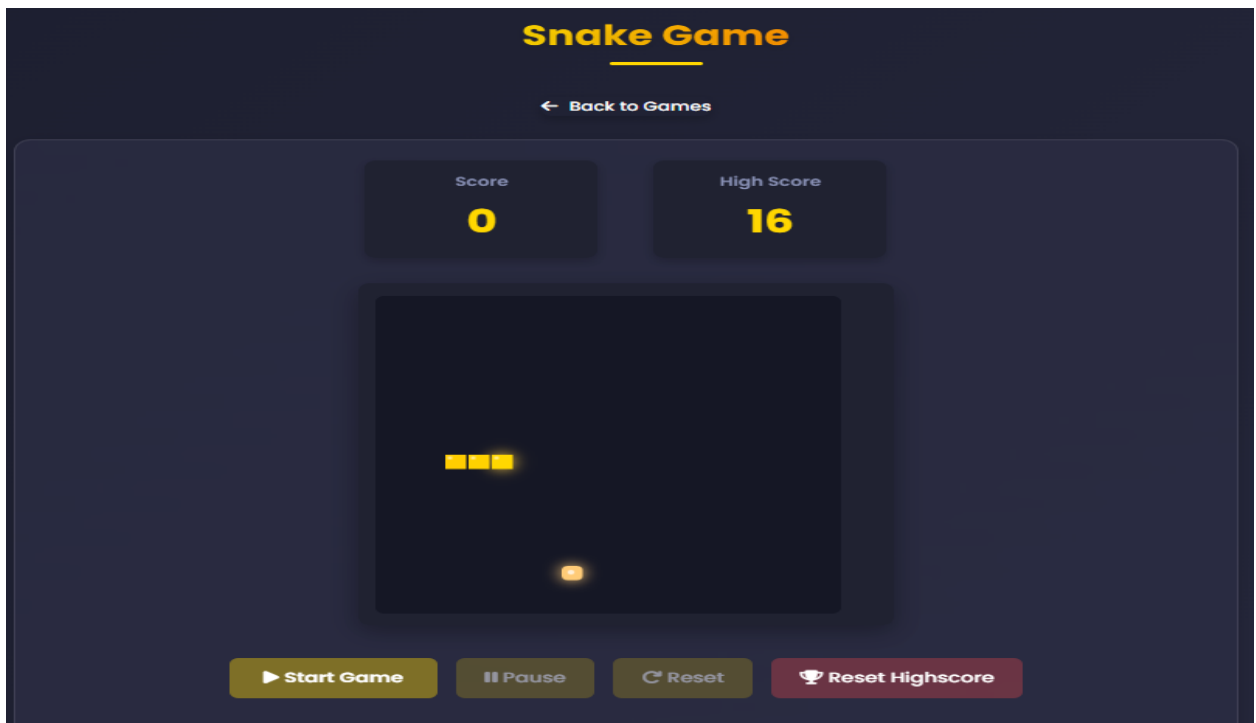


Figure 10: Snake Game

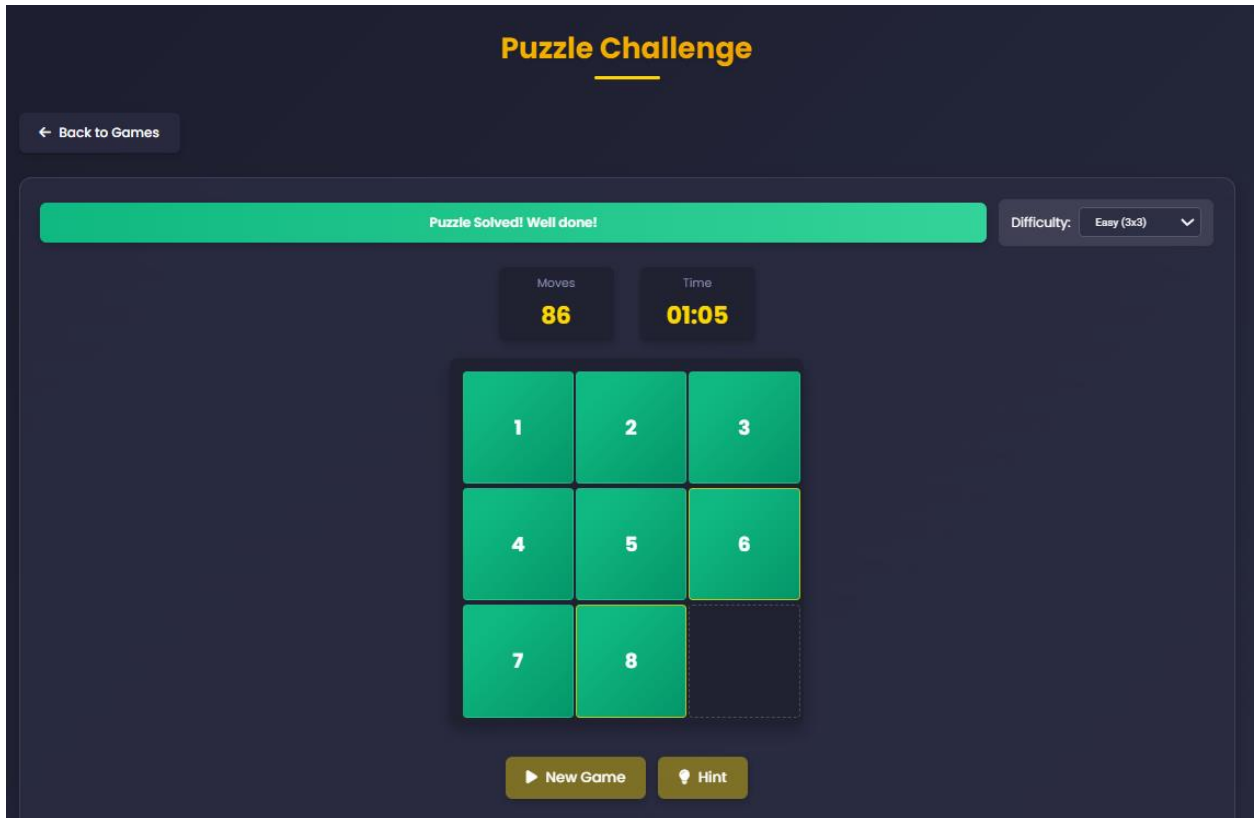


Figure 11: Puzzle Game

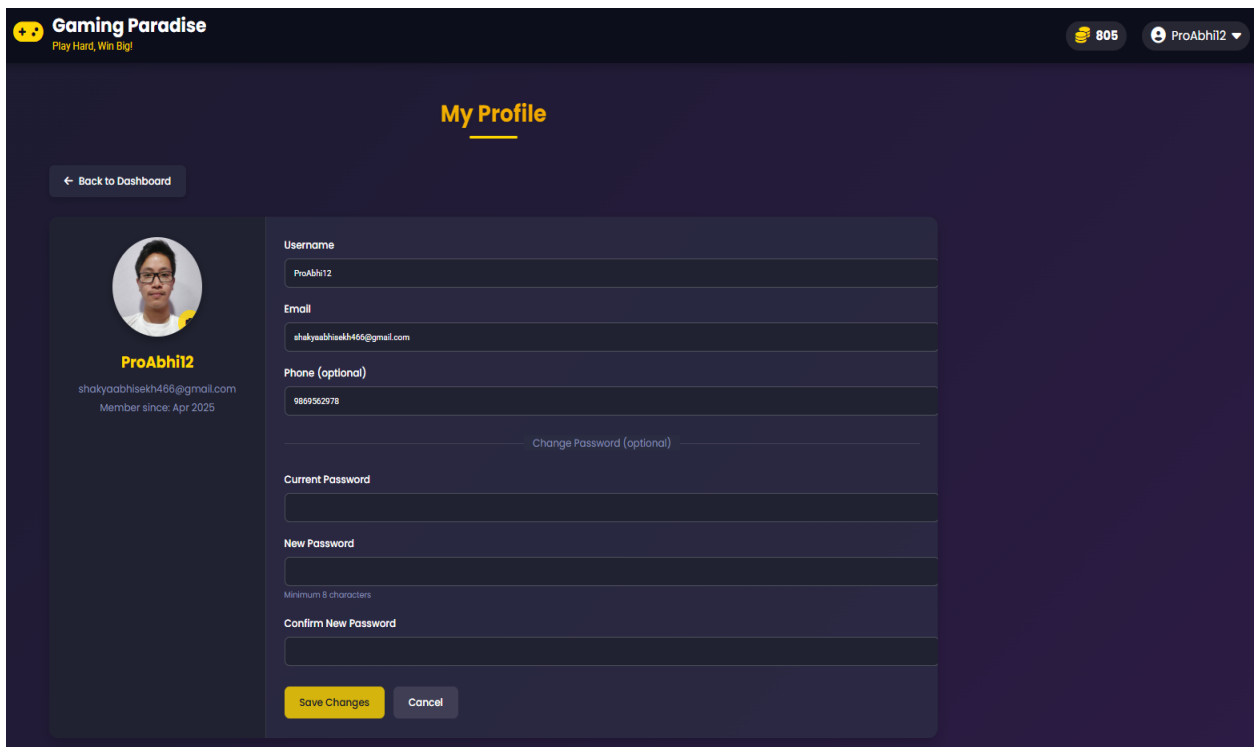


Figure 12: User Profile Page